FINAL PROJECT

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CAMPAIGN GOAL & PROPOSAL:

Goal: Total Wealth in 10 years equal to $100 million

Currently the foundation has $40 million, and to get to $100 million, the current wealth will be invested into one of seven balanced funds. These funds have different risk allocations and returns; only one fund can be chosen a year.

Plan of Action:

Create a model within R that can determine which fund the Foundation should invest in for the year. This model will simulate thousands of market paths, then select the most optimal fund based on the current wealth of the foundation while managing risk.

Maximize Profits while Considering Foundation's Risk Tolerance

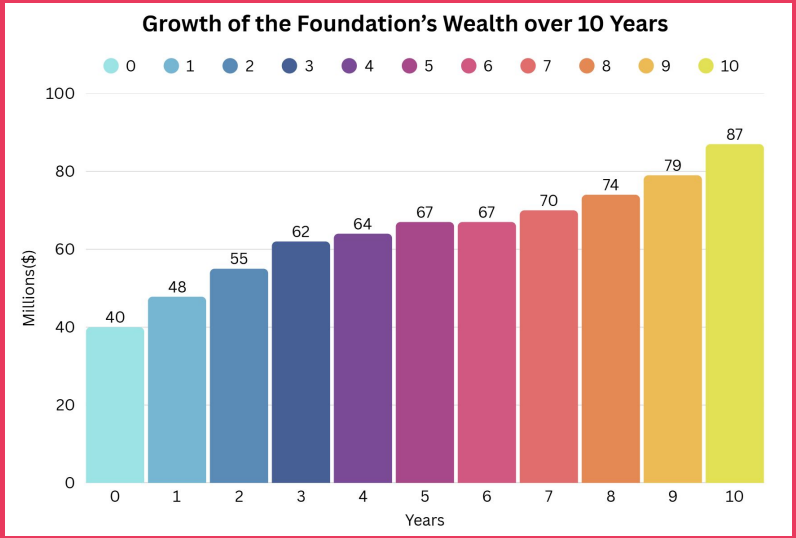
PROPOSAL RESULTS:

Key Outcomes:

Wealth: In 10 years the total projected wealth is now $87 million. With a starting wealth of $40 million, the total wealth grows by double. The standard deviation is $18 million.

Probability: The probability of reaching $100 million is 20%, and to reach $110 million it is 8%.

Funds: For the first year, invest in Fund 5, and then for the remaining years invest in Fund 4. Fund 4 is a 88% hedge fund and 12%US AGG bond, which has high return with low volatility. Fund 5 is also mostly in hedge funds (81%) with the remainder in Emerging Market Stock(16%) and US Stock(3%).



ALTERNATIVES

Is there a way for the Foundation to reach $100 million in 10 years?

1 Increase Risk Tolerance

If we increase the risk tolerance of the utility function of wealth less than 95 million. The original function is U(W) = W – 0.2(95 – W)^2, but if we lower the 0.2 multiplier then more risk can be accepted.

For example, if U(W) = W-0.01 (95-W)^2, then wealth at 10 years = $89 million. If it is lowered even more, U(W) = W-0.001(95- W)^2, then the wealth becomes $97 million, but the standard deviation is now $33 million.

2 Deadline Extension

Using the average annual growth rate of 7.7%, if the proposed deadline of 10 years is extended to 12 years then the $100 million marker to be reached.

3 Donor(s) Passing

If one of the Donors passes before the ten year marker, than an average of $5 million added to Foundation's wealth.

If one Donor passes at:

- 1 year: $94 million
- 5 year: $92 million
- 10 year: $92 million

If Both Pass at:

- 1 year: $101 million
- 5 year: $ 98 million
- 10 year: $97 million

CONCLUSIONS:

Original Plan:

A more conservative plan that doubles the initial wealth in 10 years, with a 20% chance of achieving the $100 million goal. This plan has less variance due to avoiding risky securities. If both donors pass away at the beginning of the campaign, it will be more likely to reach the $100 million goal.

Potential Changes:

1. **Extend Timeline:** Extending the timeline by around 2 years is a way to avoid increasing risk while also being able to reach the $100 million goal.
2. **Increase Risk**: While the potential for growth may grow, so does the potential for losses.

THANK YOU!